

```

4. More info at: https://en.wikipedia.org/wiki/Substitution\_cipher
5. View this code at https://nostarch.com/big-book-small-python-projects
6. Tags: short, cryptography, math"""
7.
8. import random
9.
10. try:
11.     import pyperclip # pyperclip copies text to the clipboard.
12. except ImportError:
13.     pass # If pyperclip is not installed, do nothing. It's no big deal.
14.
15. # Every possible symbol that can be encrypted/decrypted:
16. LETTERS = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'
17.
18. def main():
19.     print('Simple Substitution Cipher, by Al Sweigart
20. A simple substitution cipher has a one-to-one translation for each
21. symbol in the plaintext and each symbol in the ciphertext.')
22.
23.     # Let the user specify if they are encrypting or decrypting:
24.     while True: # Keep asking until the user enters e or d.
25.         print('Do you want to (e)ncrypt or (d)ecrypt?')
26.         response = input('> ').lower()
27.         if response.startswith('e'):
28.             myMode = 'encrypt'
29.             break
30.         elif response.startswith('d'):
31.             myMode = 'decrypt'
32.             break
33.         print('Please enter the letter e or d.')
34.
35.     # Let the user specify the key to use:
36.     while True: # Keep asking until the user enters a valid key.
37.         print('Please specify the key to use.')
38.         if myMode == 'encrypt':
39.             print('Or enter RANDOM to have one generated for you.')
40.             response = input('> ').upper()
41.             if response == 'RANDOM':
42.                 myKey = generateRandomKey()
43.                 print('The key is {}. KEEP THIS SECRET!'.format(myKey))
44.                 break
45.             else:
46.                 if checkKey(response):
47.                     myKey = response
48.                     break
49.
50.     # Let the user specify the message to encrypt/decrypt:
51.     print('Enter the message to {}'.format(myMode))
52.     myMessage = input('> ')
53.
54.     # Perform the encryption/decryption:
55.     if myMode == 'encrypt':
56.         translated = encryptMessage(myMessage, myKey)
57.     elif myMode == 'decrypt':
58.         translated = decryptMessage(myMessage, myKey)

```